DOMAIN

Natural and restricted domain, side by side

2
COMPARED

One comes from the formula alone; the other is a choice made on top of it.

	NATURAL DOMAIN	RESTRICTED DOMAIN
WHAT IT IS	the largest set of real inputs giving real output	a deliberate limitation beyond what the formula requires
SET BY	the algebraic expression alone	a purpose or the context
EXAMPLE	$\sqrt{x-3}$: $[3,\infty)$; $\ln x$: $(0,\infty)$	x^2 on $[0,\infty)$, to make it one-to-one
FROM CONTEXT	$h(t)=-16t^2+64t$ accepts any real t	launch at $t=0$, landing at $t=4$: $[0,4]$
WHEN NOTHING IS STATED	assumed: the implied domain	only if context or instruction says so

A restricted domain can never be larger than the natural one: it only removes inputs the formula would otherwise accept.